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PORTABLE CLOTHES STEAMER

Filed Feb. 27, 1968

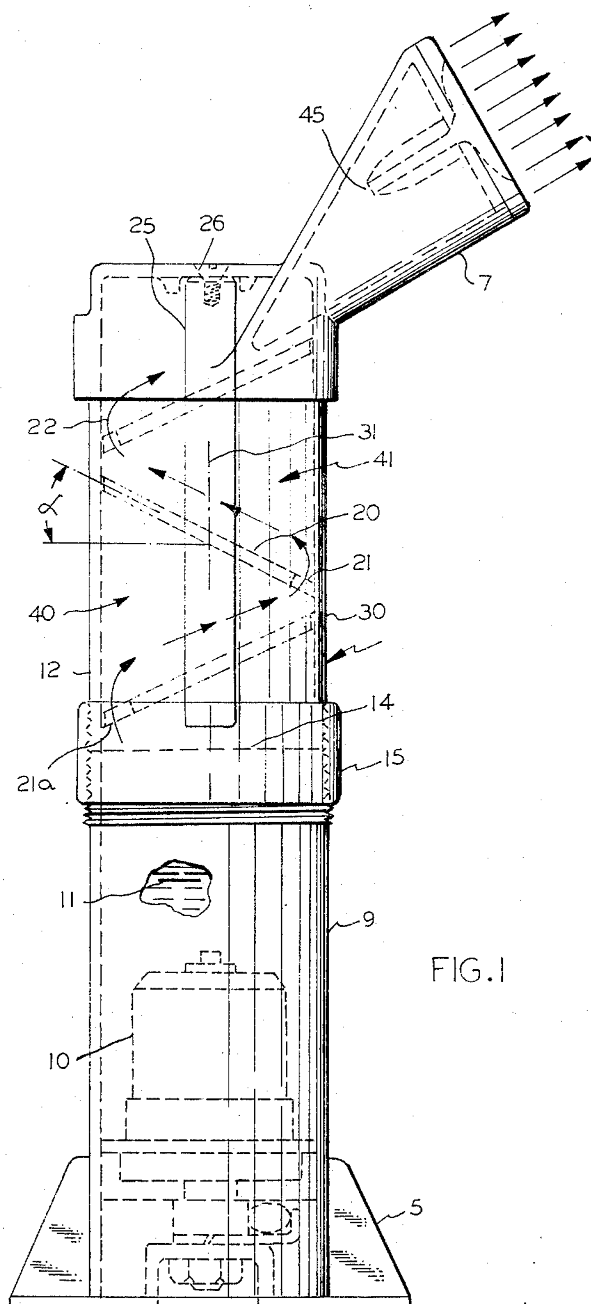

FIG.1

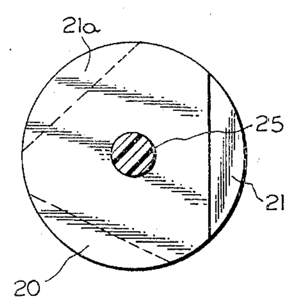

FIG.2

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PORTABLE CLOTHES STEAMER

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ABSTRACT OF THE DISCLOSURE

A housing construction is provided for a portable appliance that steams clothes to remove wrinkles. The housing comprises an uprightly disposed cylindrical type hollow watertight member with an electrically heated water chamber inside and which has upper and lower interfitting sections. A steam head is provided for treating a garment and a baffle network between is disposed in a steam conveyance path between the head and the water chamber to prevent spilling of water when the member is tilted and to prevent sputtering of steam flow by condensation of water.

This invention relates to a portable garment steamer, and more specifically, it relates to structure for conveying steam from a heated water chamber to a steam head so that it may be used in steaming clothes such as described in my copending application Ser. No. 692,828 for "Fabric Treatment Means And Methods," which is incorporated herein as a reference to the state of the prior art.

In a portable garment steamer which is manually operated and is fashioned for storage in a suitcase, problems are presented in making a watertight structure that will nevertheless generate and direct steam to a garment. If valves or shutters are used to retain water inside a steamer and to permit steam to escape, the device becomes expensive and perhaps dangerous if steam is generated and shutters or valves are not opened. Portable steamers are designed to stand in a position such that water is directed by gravity toward an internal electrical heater element. In operation, however, as the steamer is handled, it is tilted and tipped or even held in an upside down position momentarily. Also, it may be tipped over from its stand position, and in storage in a suitcase may be left for long periods in almost any position. It is desirable not to have to remove water from the device between use, and even if attempted, some drops remain that could leak.

Accordingly, it is an object of this invention to provide a simple inexpensive garment steamer.

It is another object of the invention to provide a garment steamer without any moving parts.

Still another object of this invention is to provide a garment steamer that is leakproof and which may be stored in any position with water inside.

These and other objectives and features of the invention will be recognized in the following description of a preferred embodiment of the invention, which makes reference to the accompanying drawings, wherein

FIGURE 1 is an elevation view of a garment steamer constructed in accordance with the invention, and

FIGURE 2 is a top view, partly in section, of a baffle element arrangement provided in accordance with this invention.

By reference to the drawings, it is seen that the portable garment steamer rests upon a stand **5** to hold its substantially cylindrical, hollow body **6** vertically disposed with an apertured steam head **7** at the top to permit steam **8** to escape. Basically, the body **6** is watertight with a lower water chamber **9** having an electrical heater assem-

bly **10** therein for heating the water **11**. Preferably, the body is of plastic or has some insulating holding portion if of metal, since the steamer is hand held and this insulates the hand from the heated water and steam. The body is divided into upper **12** and lower section **9** which are coupled at an intermediate position **14** by means of an interfitting coupling arrangement **15**. This coupler can be either a threaded coupler or a bayonet type removable coupling which produces a generally watertight seal at the coupling.

The upper member **12** serves as a steam conveyance means between the lower water chamber member **9** and the steam head **7**. Within this member **12** is disposed an array of baffles **20** with aperture or slot sections **21** disposed in such a manner that escaping steam must take a zig-zag path **22** out of the water chamber.

In this configuration, the top section **12** may be removed at the coupling **15** and the baffle array is affixed inside the top section by means of rod **25** and rivet or screw **26** to extend coaxially within the confines of the top section **12**. As may be seen from the top view of FIGURE 2, the rod-baffle assembly is made of a plastic material with the semi-circular wafer-like baffles **20** disposed at an angle from the vertical body axis to provide at their lowermost extremities the slot-aperture portion **21** through which the steam escapes. The baffles **20** may be of a semi-rigid plastic at least about the edges of the discs disposed on rod **25** to comprise flexible diaphragm-like contact at the engaging surfaces **30** with the interior surface of the upper section **12** when the rod-disc assembly is inserted into the top section **12**, thereby providing in essence a watertight seal except at the aperture portions **21**.

Preferably, the angle α from the vertical axis **31** at which the baffles are disposed is chosen to produces several desirable features. Any water droplets or condensation are directed by gravity back into the water chamber **9** by these baffles **20** through the reverse zig-zag path of that **22** taken in exit of the steam. This prevents any sputtering or wetting of the garment by means of water escaping with the steam **8**. Also, the number of baffles may be chosen to produce the desired valveless leakproof feature of this construction. It may be seen that the amount of water **11** is limited to a height of **14** beyond which it may not be filled when upper member **12** is removed. This is the amount that may be efficiently heated to steam and which resides below the first escape aperture **21A** when the water is heated in the stand position.

To illustrate the leakproof feature, assume that the steamer tipped on its side. The water might escape into the first compartment **40** through aperture **21A** but would remain trapped there. Slots **21**, **21A** may be staggered as shown in FIGURE 2 if desired, to assure that water does not reach the head **7** in any rotary position when the body **6** is lying on its side. When returned to upright position, the water will return to the water chamber **9**.

Even should the steamer be held temporarily in upside down position during use or in storage, the sloping baffles will constitute a series of cups designed with a capacity to hold the water capacity of water chamber **9** without reaching or spilling out of the apertures **45** in head **7**. Even should some of the water reach the head **7**, its angular disposition provides much the same action as the baffles, and the centrally disposed escape apertures **45** are surrounded by internally facing bosses or ridges so that a significant water level must be introduced to permit leakage.

Thus, it is seen that the novel structure afforded by this invention provides a simple, inexpensive, leakproof garment steamer having no moving parts or valves. Therefore, the features of novelty believed descriptive of the nature and spirit of the advance in the art provided by this invention are defined with particularity in the appended claims.

What I claim is:

1. A portable garment steamer comprising in combination, a water chamber, heating means disposed in said water chamber, an apertured steam head for treating a garment with escaping steam, conveyance means disposed between said chamber and said head comprising a hollow watertight housing coupling said chamber and said head having a removable connection dividing the steamer into two interfitting sections one of which comprises said water chamber, and a set of apertured baffles within said conveyance means directing steam escaping from the water chamber through the apertures in a zig-zag path to said head.

2. A steamer as defined in claim **1**, wherein the conveyance means and water stand together comprise substantially a hollow cylinder with a stand at the bottom end for resting the cylinder in an upright vertical position with the water chamber at the bottom, wherein the baffles are disposed at an angle from the vertical axis of the cylinder with the apertures at the lowest extremity to cause any water droplets condensing in the conveyance means to run back into the water chamber in a zig-zag path by force of gravity.

3. A steamer as defined in claim **1**, wherein the baffles are located on a member affixed to the conveyance means.

4. A steamer as defined in claim **1**, wherein said baffles are mounted integrally within said conveyance means section, extending outside the water chamber.

References Cited

UNITED STATES PATENTS

2,172,917	9/1938	Voigt	68—222
3,272,964	9/1966	Carlos et al.	219—271
3,395,469	8/1968	Gilbert	38—69

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